

Towards combined epidemiological and economic dynamical systems

This document describes some combined epi-econ models, starting with DAEDALUS, and then moving onto SFC-epi models.

Code examples are taken and adapted from <https://github.com/marcoverpas/Italy-SFC-Model> and https://github.com/marcoverpas/Six_lectures_on_sfc_models.

1 DAEDALUS

Originally designed of for the UK [Haw et al., 2022] and since applied to Indonesia [Johnson et al., 2023]. The econ model is static. The industrial sectors of the economy are used to stratify the population in the epi model, in which there are contacts related to sector of work and consumption.

In response to an outbreak, closures are mandated, resulting in operations in each sector reducing to some percentage. In the model, GVA reduces to the same percentage, and so does attendance of places of work and consumption, which scales contacts made and subsequently disease transmission.

The population is static: it does not grow, and no one changes sector of work. The non-working group remains non-working. The workforce in place includes a fraction who work from home and therefore contribute to GVA but not infections from workplace interactions.

See https://github.com/robj411/p2_drivers for a recent presentation.

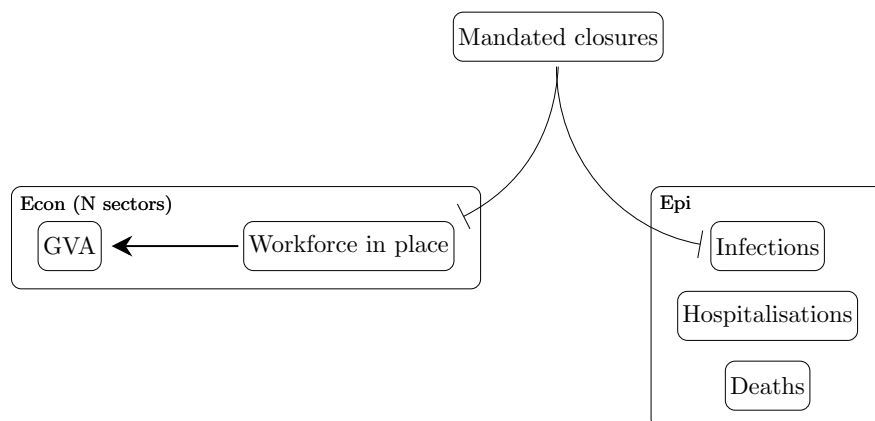


Figure 1: Daedalus model. Mandated closures reduce both infections and workforce in place.

2 SFC-epi

We will use an SFC model in place of GVA by sector. The SFC model is dynamic in time, meaning impacts can accumulate, and is demand driven, meaning we can model autonomous behaviour changes that will have impacts on both the economy and the epidemic.

At present we compute the “propensity to consume/work” which scales the contacts in the epi model and feeds into the econ model. It might be better to precompute the counterfactual (without epidemic) consumption and labour, and:

1. compute scaled parameters using epi variable(s) and response function;
2. solve econ model to get consumption and labour;
3. scale contacts with the same ratio as consumption and labour have to the no-epidemic counterfactual at the same time point

NB: the model assumes an impact on the general population (i.e. both susceptibles and infectious change their behaviour). Lack of consumption and work from people who are already sick *because* they are sick is not modelled. This could be included explicitly by e.g. setting the proportion of non-diseased people as the upper bound to propensities.

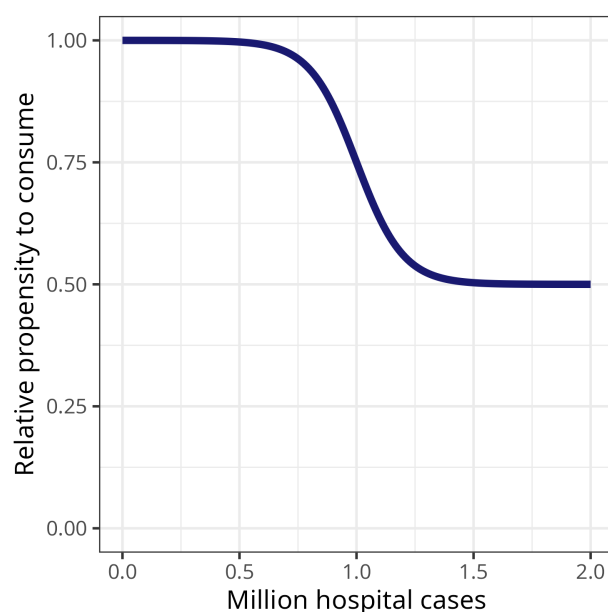


Figure 2: Dose-response function: extent of behaviour change as a function of an epidemiological variable such as number of hospital cases.

2.1 Consumption

The first step is to model consumption as a function of the epidemic, e.g. the number of cases, hospitalisations or deaths. The epidemic variable modifies the “propensity to consume” parameters (often labelled α). We scale propensity to consume and exposure-related activities by the same amount.

2.2 Labour supply

Next, we model reduction in propensity to work in the same way.

However, this should ultimately depend also on the need for income, meaning that (a) lower income people are more likely to work, (b) people are more likely to work as time goes on and they’ve spent some months with less income, and (c) we can later mitigate these effects to some extent with government transfers.

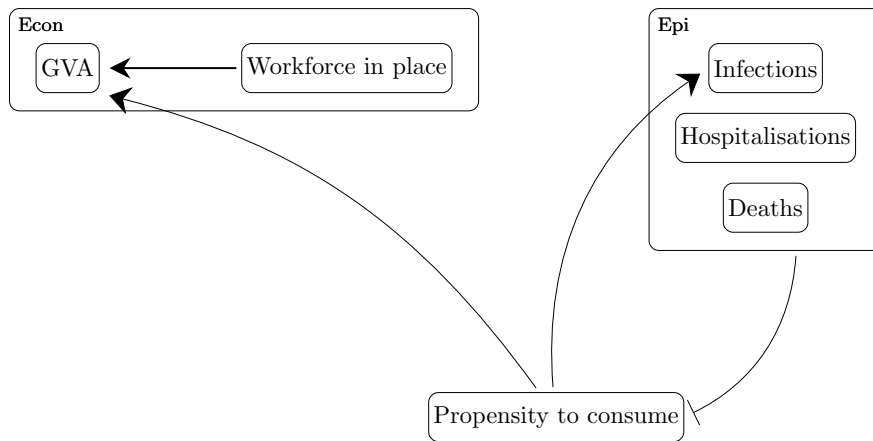


Figure 3: Epi variables reduce propensity to consume, which reduces consumption, and therefore GVA and new infections.

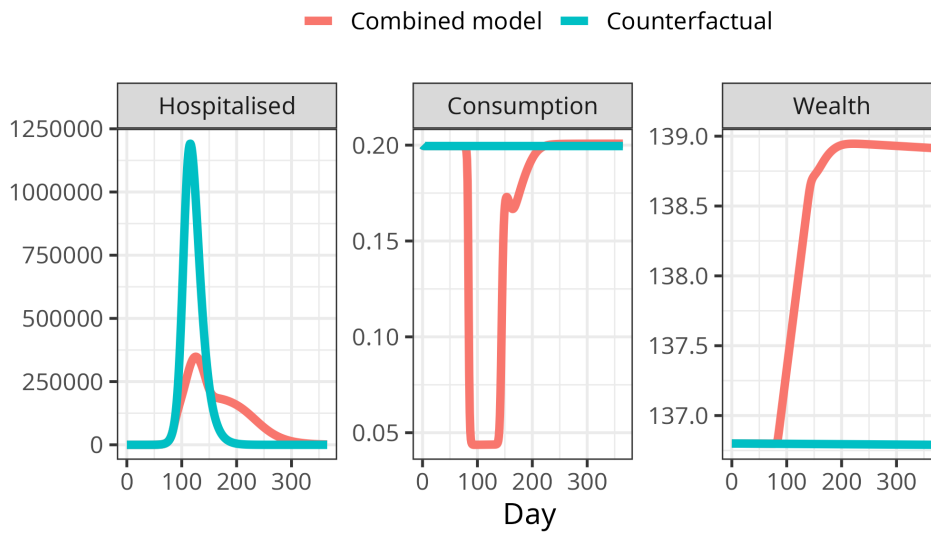


Figure 4: Results for PC model with consumption reduction alongside counterfactual epi and econ curves without integration.

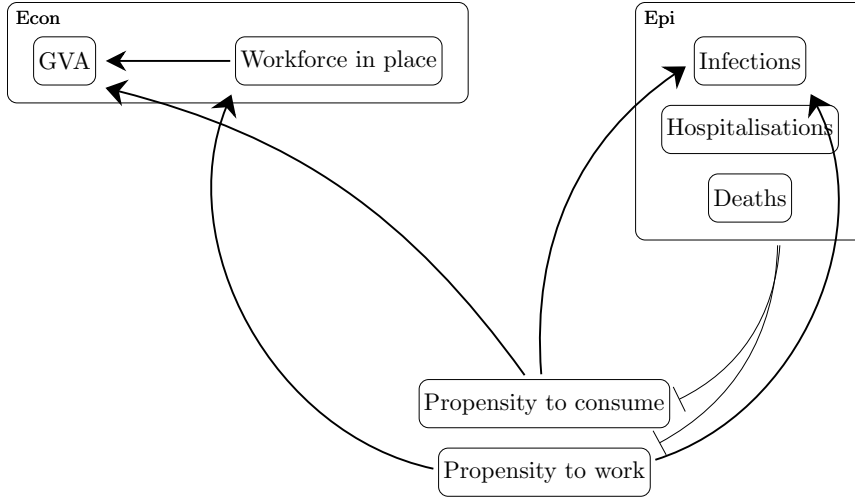


Figure 5: Epi variables reduce propensity to work and to consume, which reduces consumption and labour supply, and therefore GVA and new infections.

This version of the model (with spontaneous consumption and labour change) can be used to model an unmitigated epidemic: that is, an epidemic where there are no government interventions, but the population can choose infection-avoiding behaviours, which will dampen both the epidemic and the economy. This is an important gap in current epi modelling. We usually assume either no behaviour change, so that unmitigated epidemics have huge death tolls (e.g. UK COVID projection), or that uncostered (free) population behaviour change will curb the excesses of the epidemic (CEPI work).

2.3 Mandated closures

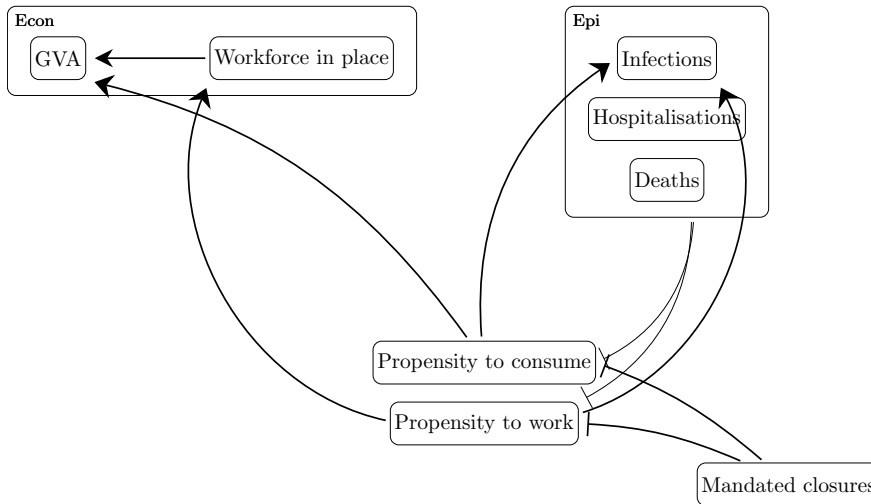


Figure 6: Epi variables reduce propensity to work and to consume, which reduces consumption and labour supply, and therefore GVA and new infections. Mandated closures reduce consumption and propensity to work.

Next we want to introduce the ability for the government to mandate closures. This might be modelled as e.g. a maximum labour demand, expressed as a percentage of counterfactual labour demand.

Propose a worked example, such as “how would we model a mandate that the hospitality sector closes?”

2.4 Government transfers

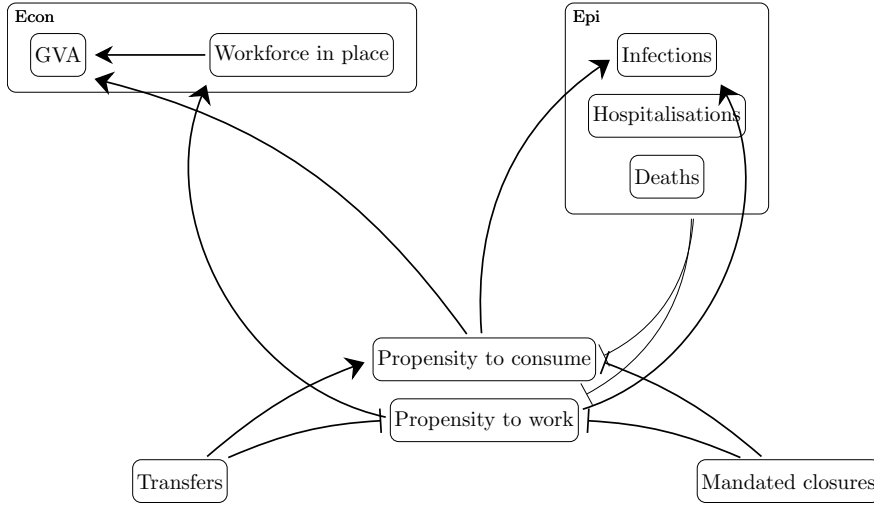


Figure 7: Epi variables reduce propensity to work and to consume, which reduces consumption and labour supply, and therefore GVA and new infections. Mandated closures reduce consumption and propensity to work. Government transfers increase consumption and reduce propensity to work.

We should introduce government transfers to demonstrate that mandated closures are only sustainable for as long as the population is supported to forego income in order to stop the spread of infection.

2.5 Other (important) things to consider

- international trade, esp. tourism
- structural changes over time?
- move to online consumption

3 Epidemic model

I propose that, for the epidemic model, we remain as consistent as possible with DAEDALUS:

- four age groups (pre-school age, school-age children, working age, retirement age)
- working-age people stratified by something: definitely working/not working, potentially something sector, occupation or SES related
- seven disease states (S, E, I (symptomatic/asymptomatic), R, H, D)
- no vaccination or waning, for simplicity
- leave school closures out for now, for simplicity
- outputs from the econ model parametrise the function k_j^1

3.1 Ordinary differential equations

$$\frac{dS_j}{dt} = -k_j^1(t)S_j \quad (1)$$

$$\frac{dE_j}{dt} = k_j^1(t)S_j - (k^2 + k^4)E_j \quad (2)$$

$$\frac{dI_j^a}{dt} = k^2E_j - k^3I_j^a \quad (3)$$

$$\frac{dI_j^s}{dt} = k^4E_j - (k_j^5 + k_j^6)I_j^s \quad (4)$$

$$\frac{dR_j}{dt} = k^3I_j^a + k_j^5I_j^s + k_j^7(t)H_j \quad (5)$$

$$\frac{dH_j}{dt} = k_j^6I_j^s - (k_j^7(t) + k_j^8(t))H_j \quad (6)$$

$$\frac{dD_j}{dt} = k_j^8(t)H_j \quad (7)$$

3.2 Disease state transitions

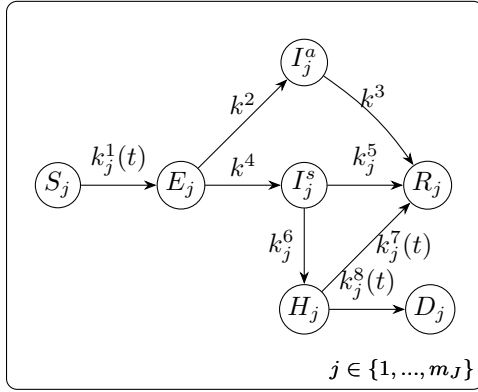


Figure 8: Disease state transitions. S : susceptible. E : exposed. I^a : asymptomatic infectious. I^s : symptomatic infectious. H : in need of hospitalisation. R : recovered. D : died. j : stratum.

References

- David Haw, Giovanni Forchini, Patrick Doohan, Paula Christen, Matteo Pianella, Rob Johnson, Sumali Bajaj, Alexandra B Hogan, Peter Winskill, Marisa Miraldo, Peter J White, Azra C Ghani, Neil M Ferguson, Peter C Smith, and Katharina Hauck. Optimizing social and economic activity while containing SARS-CoV-2 transmission using DAEDALUS. *Nature Computational Science*, 2:223–233, 2022. doi: 10.1038/s43588-022-00233-0.
- Rob Johnson, Bimandra Djaafara, David Haw, Patrick Doohan, Giovanni Forchini, Matteo Pianella, Neil Ferguson, Peter C Smith, and Katharina D Hauck. The societal value of SARS-CoV-2 booster vaccination in Indonesia. *Vaccine*, (41), 2023. ISSN 0264-410X. doi: 10.1016/j.vaccine.2023.01.068.